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***Abstract*—Early and accurate detection of eye diseases is critical to preventing irreversible vision loss, yet most deep learning diagnostic models remain limited by their "black-box" nature and lack of quantitative interpretability validation. This study presents a comprehensive comparative evaluation between modern Convolutional Neural Networks (ConvNeXt-Tiny) and Vision Transformers (DeiT-Small) for multiclass retinal disease classification using fundus images (categorized into cataract, diabetic retinopathy, glaucoma, and normal classes). Beyond standard classification performance metrics—where DeiT-Small and ConvNeXt-Tiny achieved comparable overall macro-F1 scores of 87.74% and 87.63%, respectively—this work provides a rigorous quantitative assessment of visual explanation faithfulness. Specifically, we evaluate the explanation fidelity of Grad-CAM (for ConvNeXt) and Attention Rollout (for DeiT) using patch-based deletion and insertion Area Under the Curve (AUC) metrics. Our findings demonstrate that while both architectures exhibit strong discriminative power and clinical alignment on localized lesions (such as diabetic retinopathy and glaucoma), their explainability profiles differ significantly in spatial focus and faithfulness. This research advances medical image analysis by bridging the gap between classification accuracy and explainability validation, offering empirical guidelines for selecting optimal model-XAI combinations in clinical decision-support systems.**

***Index Terms*—VLM, surveillance, LoRA, DoRA**

I. Introduction

Vision impairment affects over 2.2 billion people globally, mainly due to cataracts, diabetic retinopathy, and glaucoma [1]. Cataracts cause 45% of global blindness (15.2 million people) [2], diabetic retinopathy is projected to reach 160.50 million cases by 2045 [3], and glaucoma, the leading cause of irreversible blindness, is expected to affect 111.8 million people by 2040 [4]. These figures underline the urgent need for scalable early detection, especially in low- and middle-income regions with limited eye care access.

Deep learning, particularly Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs), has become foundational in medical image analysis by extracting features from ophthalmic images without manual engineering [5]. ConvNeXt modernizes CNNs with Transformer design principles to improve accuracy while preserving efficiency [6], whereas DeiT uses knowledge distillation to train ViTs effectively on smaller datasets at lower computational cost [7].

Despite their high accuracy, deep learning models remain black boxes whose decision-making is difficult for humans to interpret [8]. Explainable Artificial Intelligence (XAI) addresses this by visualizing the reasoning behind predictions: Grad-CAM produces importance heatmaps for CNNs such as ConvNeXt, while Attention Rollout or Gradient Attention Rollout aggregates layer-wise attention weights in Transformers such as DeiT [9]. Because these mechanisms differ, their visualizations are not directly comparable, yet most medical deep learning evaluations stop at accuracy. Faithfulness can

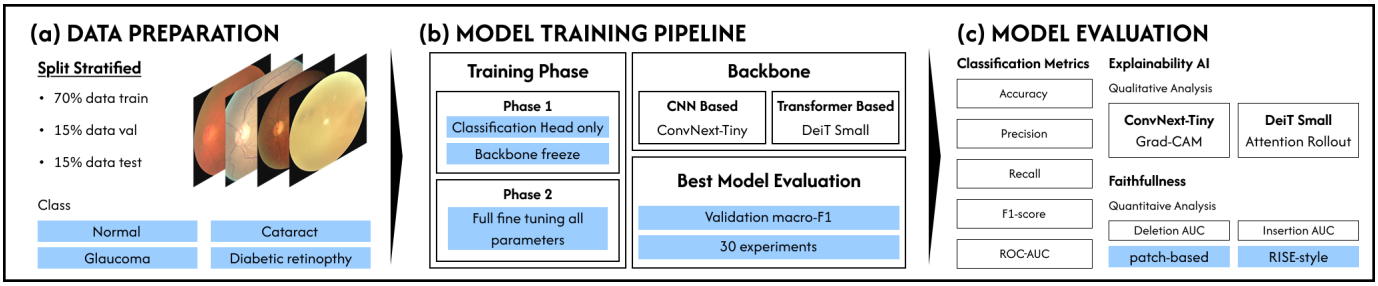


Fig. 1. Overview of the research methodology.

instead be quantified with deletion and insertion metrics, which progressively remove or reveal image regions by importance and measure the resulting change in model confidence as an Area Under the Curve (AUC) [8].

Accordingly, this research has two analytical focuses. First, it compares the classification performance of ConvNeXt and DeiT on a four-class eye disease dataset (cataract, diabetic retinopathy, glaucoma, and normal) using accuracy, precision, recall, and F1-score. Second, it compares the faithfulness of Grad-CAM on ConvNeXt and Attention Rollout on DeiT using deletion and insertion AUC.

This research offers two contributions. Theoretically, it enriches the literature on the comparative interpretability of modern CNN (ConvNeXt) and ViT (DeiT) architectures in ophthalmic imaging by evaluating models on the faithfulness of their XAI visualizations rather than accuracy alone, an aspect rarely explored in previous studies. Practically, it provides empirical guidance for clinicians and engineers in choosing the architecture–XAI combination best suited to operational systems, particularly when a diagnostic system requires heatmaps closely aligned with the model's decision basis.

II. Related Work

A. Deep Learning-Based Eye Disease Classification

Deep learning-based eye disease classification using fundus images has attracted significant research attention in recent years. The Eye Disease Classification dataset [10] is one of the publicly available datasets frequently used for benchmarking multiclass eye disease classification models. The characteristics of the dataset are presented in Table I.

TABLE I
DISTRIBUTION OF EYE DISEASE CLASSIFICATION DATASET

Class	Number of Images	Percentage (%)
Cataract	1,038	24.62
Diabetic Retinopathy	1,098	26.04
Glaucoma	1,007	23.88
Normal	1,074	25.47
Total	**4,217**	**100.00**

Dataset characteristics are relatively consistent across studies, comprising 4,217 fundus images categorized into four classes: cataract, diabetic retinopathy, glaucoma, and normal eyes [11], [12]. Although the dataset is relatively balanced, glaucoma represents the minority class, which motivates the use of data augmentation techniques during preprocessing. Furthermore, the dataset provides inherent inter-subject variability across different patients. Similar cross-subject challenges in biomedical data were highlighted by Colot et al. [13], emphasizing the need for robust feature evaluation beyond standard accuracy.

The dataset has been utilized in several studies to evaluate the performance of deep learning models. A comparative study involving DenseNet, EfficientNet, Xception, VGG, and ResNet was conducted in [11]. Among these architectures, EfficientNet achieved the highest accuracy of 87.84%. Subsequently, transfer learning with EfficientNet-B3 combined with data augmentation and cosine annealing learning rate scheduling was proposed in [14], resulting in an improved accuracy of 95.12%.

A transfer learning framework named TL-MED was introduced in [12], where five CNN-based architectures and an ensemble model were evaluated. DenseNet169 achieved the highest individual accuracy of 96%. In another study, AlexNet, EfficientNet-B0, EfficientNet-B7, and a custom CNN architecture were compared [15]. AlexNet demonstrated the best performance with an accuracy of 93.65%, and the study further employed SHAP-based interpretability analysis to identify clinically relevant image regions. Beyond SHAP, gradient-based visualization techniques such as Grad-CAM [16] have also been widely adopted in fundus image classification studies to highlight discriminative regions contributing to the model's prediction, providing a complementary means of interpreting CNN-based decisions.

Based on these studies, existing research on this dataset primarily focuses on conventional CNN architectures and their variants, including VGG, ResNet, DenseNet, EfficientNet, and AlexNet. In contrast, investigations involving modern CNN architectures such as ConvNeXt and transformer-based architectures such as DeiT remain limited. This research gap motivates the comparative analysis conducted in this study.

B. ConvNeXt and DeiT Architectures for Image Analysis

As representative architectures of modern convolutional networks and vision transformers, ConvNeXt and DeiT have demonstrated promising performance across various image classification tasks. ConvNeXt was introduced in [6] as a modernization of the ResNet architecture by incorporating several design principles inspired by Vision Transformers, including patchify stems, depthwise convolutions, inverted bottlenecks, larger kernel sizes, and simplified normalization and activation layers.

On the other hand, DeiT (Data-efficient Image Transformer) was proposed in [7] as an extension of the original Vision Transformer architecture [17]. Since the attention mechanism on its own does not take the order of tokens into consideration, positional encodings are integrated into the patch embeddings [18]. DeiT introduces a knowledge distillation strategy that enables transformer models to be trained effectively on relatively smaller datasets while maintaining competitive performance. Beyond classification, the adaptation of transformer architectures and attention mechanisms has also demonstrated strong efficacy across diverse computer vision domain tasks [19].

Several comparative studies have investigated the strengths and limitations of CNN and Vision Transformer architectures in image analysis tasks. A literature review presented in [20] concluded that CNNs benefit from strong inductive biases and superior data efficiency, whereas Vision Transformers are more effective at capturing global contextual information but

generally require larger datasets and greater computational resources.

In the specific domain of fundus image analysis, a comparative study between CNN-based architectures, including ConvNeXt, and transformer-based models was conducted in [21]. The results showed that the transformer-based CaiT architecture achieved the highest AUC of 0.979, although the performance gap between the best CNN and transformer models was relatively small. Similar findings were reported in [22], where Vision Transformer models achieved non-inferior or superior performance compared with CNNs on five out of six glaucoma datasets.

Overall, these findings suggest that the relative advantages of modern CNN architectures such as ConvNeXt and transformer-based architectures such as DeiT depend heavily on dataset characteristics and training strategies. Therefore, no clear consensus has emerged regarding the superiority of either approach. This motivates a direct comparison between ConvNeXt and DeiT for multiclass eye disease classification using the Eye Disease Classification dataset.

III. Methodology

A. Model Architecture

1) ConvNeXt: ConvNeXt is a modern convolutional neural network (CNN) architecture inspired by Vision Transformers (ViTs). It modernizes standard convolutional operations without relying on self-attention mechanisms. Specifically, ConvNeXt redesigns the standard ResNet architecture by adopting 7×7 depthwise convolution to enlarge the receptive field, using inverted bottleneck blocks, replacing Batch Normalization with LayerNorm, and replacing Rectified Linear Unit (ReLU) with GELU to improve optimization stability during transfer learning [6]. These architectural refinements enhance feature representation without sacrificing the computational advantages of convolution-based networks. Fig. 2 illustrates the internal structural flow of a ConvNeXt block.

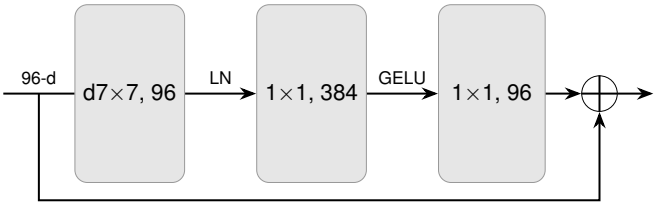


Fig. 2. ConvNeXt Block Architecture.

To evaluate its real-world efficacy, a recent large-scale study compared ConvNeXt, RegNet, and ResNet across 22 publicly available retinal fundus datasets [23]. This benchmark comprised more than $125,000$ images spanning normal cases, diabetic retinopathy, glaucoma, and age-related macular degeneration to construct a clinically representative screening model. Among the tested architectures, ConvNeXt-Tiny achieved the best overall performance across most evaluation metrics, obtaining an overall accuracy of $80.46 \pm 1.48\%$. It demonstrated particularly high disease recognition accuracy for glaucoma

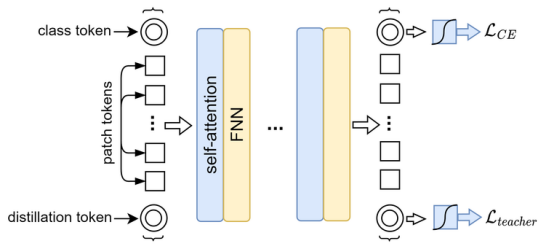


Fig. 3. DeiT Block Architecture

(97.20%), age-related macular degeneration (98.14%), and diabetic retinopathy (80.66%) These findings demonstrated ConvNeXt-Tiny's reliability for retinal disease classification while maintaining a relatively lightweight architecture, making it highly suitable for transfer learning on moderately sized retinal fundus datasets.

2) DeiT: Standard Vision Transformers (ViTs) rely primarily on self-attention mechanisms to capture long-range dependencies across an entire image and generally require extremely large-scale pretraining datasets to achieve competitive performance [17]. To overcome these substantial data requirements while preserving the advantages of transformer-based architectures, Data-efficient Image Transformers (DeiT) were proposed [7]. DeiT introduces a dedicated distillation token into the architecture, which is appended alongside the class token and patch embeddings as illustrated in Fig. 3. During training, this token interacts with the patch tokens via self-attention to learn complementary features from a robust, pretrained teacher model (typically a CNN). This mechanism significantly improves classification accuracy, particularly for smaller transformer models, while maintaining computational efficiency.

The data-efficient design of DeiT makes it particularly suitable for medical image analysis, where large annotated datasets are often unavailable and training standard ViTs from scratch is difficult. A recent study incorporated DeiT as one of the transformer-based feature extractors in a hybrid framework for multi-class eye disease classification, demonstrating that combining DeiT with machine learning classifiers contributed to a stacking model that achieved 97.163% classification accuracy on a benchmark retinal dataset [24]. These findings indicate that transformer-based models can be effectively applied to retinal disease classification when equipped with efficient training or feature extraction strategies. Therefore, DeiT is suitable for transfer learning as a representative data-efficient Vision Transformer for a moderately sized retinal fundus datasets.

3) Hyperparameter & Model Selection: Hyperparameters are user-defined settings that control the training process, whereas model parameters are the network weights automatically learned through backpropagation [25]. Both ConvNeXt and DeiT were fine-tuned using the AdamW optimizer [26] with a cosine learning-rate decay schedule, following the

official training recipes of both architectures. Although the original papers employ different optimization settings, particularly for large-scale ImageNet pretraining, reproducing their exact recipes was impractical given the hardware constraints and the smaller size of the diabetic retinopathy dataset. A unified configuration was therefore adopted for both models to ensure a fair comparison.

Rather than exhaustively tuning every hyperparameter, only the fine-tuning learning rate and the fine-tuning epoch budget were optimized, using a grid search over three learning rates (10^{-4}, 5×10^{-5}, and 10^{-5}) and five epoch budgets (10, 20, 30, 40, and 50), giving 15 configurations per architecture. Restricting the search to these two hyperparameters follows Bergstra and Bengio [27], who showed that only a small subset of hyperparameters typically has a significant impact on model performance, which keeps an exhaustive grid over the remaining search space computationally tractable. All remaining settings were fixed across both architectures and selected from their official training recipes or adjusted to the available computational budget. For each architecture, the configuration with the highest validation macro-F1 was selected. Table II summarizes the final configuration, where values that differ between the two models are given as ConvNeXt / DeiT.

TABLE II
FINAL HYPERPARAMETER CONFIGURATION.

Hyperparameter	Value	Source / Justification
Optimizer	AdamW	Common optimizer for both models
LR scheduler	Cosine decay	Adopted from both models' official recipes
Weight decay	0.05	Adopted from DeiT training recipe and applied consistently to both models
Label smoothing	0.1	Recommended by both ConvNeXt and DeiT papers
Linear probe LR	10^{-3}	Empirically selected for the linear probing stage
Linear probe epochs	5	Chosen to initialize the classifier before fine-tuning
Fine-tuning epochs	30 / 40	Grid search over $\{10, 20, 30, 40, 50\}$
Fine-tuning LR	10^{-5} / 10^{-4}	Grid search over $\{10^{-4}, 5 \times 10^{-5}, 10^{-5}\}$
Batch size	32	Limited by available GPU memory

To ensure an unbiased evaluation, the dataset is divided into training, validation, and testing subsets. the training set optimized the model parameters, the validation set selected the best hyperparameter configuration [28], and the test set was reserved solely for final performance evaluation, without involvement in any stage of model selection [29]. This separation prevents information leakage and optimistic bias, yielding a more reliable estimate of the model's generalization capability.

B. Explainable AI (XAI) on Computer Vision

1) Grad-CAM: For a target class c, Grad-CAM first backpropagates the class score y^c with respect to each feature map A^k in the final convolutional layer. The importance weight of the k-th feature map is obtained by applying global average pooling to the spatial gradients, which is defined as

$$\alpha_k^c = \frac{1}{Z} \sum_i \sum_j \frac{\partial y^c}{\partial A_{i,j}^k} \tag{1}$$

where Z denotes the total number of spatial locations in the feature map. The resulting weight α_k^c represents the contribution of feature map A^k to the prediction of class c [30].

The class activation map is subsequently generated as a weighted combination of all feature maps,

$$L_{Grad-CAM}^c = ReLU(\sum_k \alpha_k^c A^k) \tag{2}$$

where the ReLU suppresses negative activations, preserving only features that positively influence the prediction of the target class [30].

Although originally developed for CNNs, Grad-CAM has been successfully adapted to Vision Transformers. The method utilizes the patch token representations extracted from the final transformer encoder block, which are reshaped into a two-dimensional spatial grid before gradient computation. The spatial gradients are globally averaged to obtain token importance weights, followed by weighted aggregation and ReLU activation to produce the final saliency map [31], [32].

Grad-CAM has been widely adopted in medical image analysis to verify whether deep learning models focus on clinically relevant anatomical structures or pathological lesions. As summarized in Table III, Grad-CAM has demonstrated its effectiveness across multiple imaging modalities and deep learning architectures.

TABLE III
APPLICATIONS OF GRAD-CAM IN MEDICAL IMAGE ANALYSIS.

Medical Domain	Architecture	Interpretability Outcome
Chest X-ray (TB)	ViT	Highlights diagnostically relevant pulmonary regions, enabling visualization of the model's focus during tuberculosis classification [32].
Retinal disease	ResNet101-V2	Emphasizes retinal abnormalities associated with glaucoma, diabetic retinopathy, and cataracts, improving multiclass model interpretability [33].
Thyroid ultrasound	Hybrid CNN–ViT	Produces clinically meaningful activation patterns that validate lesion localization and support prediction reliability [31].

2) Attention Rollout: Vision Transformers compute a self-attention matrix at every transformer layer, where each element represents the attention weight assigned from one token to another. Since each image patch is represented as a token, the attention matrix describes how information is exchanged among all image patches through the self-attention mechanism. Self-attention enables every token to interact with all other tokens, allowing the model to capture global contextual relationships, unlike convolutional neural networks that rely on local receptive fields [17].

However, the attention matrix from a single transformer layer only reflects the interaction at that particular layer and does not directly indicate the overall contribution of an input patch to the final prediction. To estimate how information propagates throughout the transformer, Attention Rollout recursively composes the attention matrices from consecutive layers. Consequently, the importance of each input patch is determined by the accumulated attention flow from the input layer to the final class token representation [34].

The recursive formulation of Attention Rollout is expressed as

$$\tilde{A}(l_i) = \begin{cases} A(l_i)\tilde{A}(l_{i-1}) & \text{if } i > j \\ A(l_i) & \text{if } i = j \end{cases} \tag{3}$$

where $A(l_i)$ denotes the attention matrix of the i-th transformer layer, while $\tilde{A}(l_i)$ represents the accumulated attention after recursively composing the attention matrices from previous layers. For image classification, the attention values corresponding to the class (CLS) token in the final rollout matrix are reshaped into the spatial arrangement of the input patches to generate the explanation heatmap [34].

Compared with gradient-based explanation methods, Attention Rollout provides a global interpretation by tracing information flow through all transformer layers rather than relying on the gradients of a single layer. The resulting heatmaps tend to be more diffused and less spatially localized than those produced by Grad-CAM, because the explanation aggregates attention across multiple layers [35], [36]. A summary of the differences between Attention Rollout and Grad-CAM is presented in Table IV.

TABLE IV
COMPARISON BETWEEN GRAD-CAM AND ATTENTION ROLLOUT

Method	**Characteristics**
Grad-CAM	Produces class-discriminative heatmaps using gradients from the last convolutional feature map. It provides relatively sharp and localized visual explanations but is primarily designed for CNNs.
Attention Rollout	Propagates attention recursively across transformer layers to estimate the contribution of each input patch to the CLS token. It captures global contextual relationships but generally produces more diffuse heatmaps than Grad-CAM.

C. Faithfulness Evaluation

The quality of the generated saliency maps was quantitatively evaluated using the deletion and insertion Area Under the Curve (AUC) metrics. These metrics are both model-agnostic and explanation-agnostic because they assess explanation quality based solely on changes in the model prediction after perturbing the input image, regardless of the internal mechanism used to generate the explanation [37], [38]. Deletion AUC measures whether the regions identified as important by a saliency map are truly critical for the model prediction, while insertion AUC measures how effectively the highlighted regions restore the model prediction.

The original formulation perturbs the image pixel by pixel [37]. In this study, the perturbation is instead applied at the patch level: every heatmap is expressed on a common 14×14 grid of 16×16-pixel patches over the 224×224 input, which is the native patch grid of DeiT, and the Grad-CAM map of ConvNeXt is bilinearly resized to the same grid. Using the same perturbation granularity for both architectures keeps their faithfulness scores directly comparable. As the uninformative baseline, a heavily Gaussian-blurred copy of the input image (kernel size 51, $\sigma = 41$) is used rather than a constant black or gray image, so that perturbed regions do not introduce sharp artificial edges that the model has never seen during training.

Deletion AUC starts by ranking all 196 patches according to their importance values in the generated heatmap. The most important patches are then progressively replaced, one patch per step, with the corresponding region of the blurred baseline while the softmax confidence of the predicted class is recorded after each step. The AUC is then calculated with the trapezoidal rule from the resulting confidence values plotted against the fraction of removed patches, where a lower AUC indicates a more faithful explanation because the model confidence decreases rapidly when the truly relevant regions are removed [35].

Insertion AUC starts from the blurred baseline image, and the original patches are gradually inserted in the same order of importance. The AUC is then calculated from the prediction confidence plotted against the fraction of inserted patches, where a higher AUC indicates a more faithful explanation because the selected patches rapidly recover the confidence of the target prediction [35].

IV. RESULTS

This section presents the experimental results of the study, organized according to the two analytical scopes defined in Section I: (1) the classification performance comparison between ConvNeXt-Tiny and DeiT-Small, and (2) the faithfulness comparison between Grad-CAM and Attention Rollout as measured by deletion and insertion AUC.

A. Hyperparameter Selection

Both ConvNeXt-Tiny and DeiT-Small were fine-tuned using a two-phase training scheme (linear-probe followed by full fine-tuning) under a grid search over learning rate and number of training epochs. For each architecture, the configuration

achieving the highest validation macro-F1 was selected as the final model, following the model selection protocol described in Section III-A. Table V summarizes the selected configuration for each architecture.

TABLE V
SELECTED HYPERPARAMETER CONFIGURATION PER ARCHITECTURE

Hyperparameter	ConvNeXt-Tiny	DeiT-Small
Linear probe learning rate	1×10^{-3}	1×10^{-3}
Full fine-tuning learning rate	1×10^{-5}	1×10^{-4}
Weight decay	0.05	0.05
Label smoothing	0.1	0.1
Linear probe epochs	5	5
Full fine-tuning epoch budget	30	40
Batch size	32	32

Table VI reports the validation macro-F1 of all 15 grid-search configurations per architecture. The scores vary by less than two percentage points within each architecture (89.78–91.48% for ConvNeXt-Tiny and 90.37–92.34% for DeiT-Small), indicating that neither model is highly sensitive to these two hyperparameters. ConvNeXt-Tiny performed best with the smallest learning rate, whereas DeiT-Small benefited from the largest learning rate combined with a longer epoch budget.

TABLE VI
VALIDATION MACRO-F1 (%) OF ALL GRID-SEARCH CONFIGURATIONS

		Full fine-tuning epoch budget				
Model	**LR**	**10**	**20**	**30**	**40**	**50**
ConvNeXt-Tiny	1×10^{-5}	91.30	91.32	**91.48**	91.48	91.33
	5×10^{-5}	90.51	91.01	90.90	91.02	91.35
	1×10^{-4}	91.32	90.24	89.78	90.65	90.56
DeiT-Small	1×10^{-5}	90.37	91.35	91.51	91.83	92.30
	5×10^{-5}	91.48	91.63	92.13	91.97	91.61
	1×10^{-4}	91.33	91.87	92.01	**92.34**	92.15

The selected configuration for each architecture reached its best validation macro-F1 well before its epoch budget was exhausted (fine-tuning epoch 7 of 30 for ConvNeXt-Tiny and 15 of 40 for DeiT-Small).

B. Classification Performance

Table VII presents the test-set classification performance of both architectures, reported as overall accuracy and macro-averaged precision, recall, and F1-score. Diabetic retinopathy recall is reported separately, as it serves as the tie-breaker criterion for final model selection due to the higher clinical cost of false-negative diagnoses.

Table VIII further reports the per-class precision (P), recall (R), and F1-score of both models, providing a more detailed view of class-wise classification behavior, particularly for the minority glaucoma class.

Both architectures reached an identical test accuracy of 87.99%, each correctly classifying 557 of the 633 test images, so accuracy alone cannot separate them. Based on the test-set macro-F1, DeiT-Small (87.74%) marginally outperformed ConvNeXt-Tiny (87.63%) – a gap of only 0.11 percentage points, small enough that the two architectures are effectively comparable in overall discriminative power. Since the macro-F1 values were not exactly tied, the diabetic retinopathy recall tie-breaker was not formally invoked; however, it is worth noting that ConvNeXt-Tiny attained a substantially higher DR recall (99.39% vs. 95.76%) and macro ROC-AUC (98.15% vs. 97.06%), a discrepancy discussed further in Section IV-E. Per the strict macro-F1 ranking, DeiT-Small is treated as the best-performing architecture; both architectures' checkpoints are nonetheless carried forward into the explainability analysis in the following subsections, since faithfulness is compared architecture-to-architecture (Grad-CAM vs. Attention Rollout) rather than reduced to a single winner.

TABLE VII
TEST-SET CLASSIFICATION PERFORMANCE

Metric	ConvNeXt-Tiny	DeiT-Small
Accuracy	87.99%	87.99%
Macro Precision	87.91%	88.03%
Macro Recall	87.74%	87.77%
Macro F1-score	87.63%	**87.74%**
Macro ROC-AUC	**98.15%**	97.06%
DR Recall (tie-breaker)	**99.39%**	95.76%

TABLE VIII
PER-CLASS TEST-SET PERFORMANCE OF BOTH MODELS (%)

	ConvNeXt-Tiny			DeiT-Small		
Class	**P**	**R**	**F1**	**P**	**R**	**F1**
Cataract	91.08	91.67	91.37	93.46	91.67	92.56
DR	87.70	99.39	93.18	87.78	95.76	91.59
Glaucoma	85.50	74.17	79.43	86.26	74.83	80.14
Normal	87.34	85.71	86.52	84.62	88.82	86.67

The per-class results in Table VIII show where the two models differ despite their equal accuracy. The two models share the same cataract recall (91.67%), while DeiT-Small is stronger on the normal class (88.82% vs. 85.71% recall) and ConvNeXt-Tiny is stronger on diabetic retinopathy (99.39% vs. 95.76% recall). Both models show the same weak point: glaucoma has the lowest recall of the four classes (74.17% for ConvNeXt-Tiny and 74.83% for DeiT-Small), and it is also the smallest class in the test set (151 images).

Fig. 4 shows the confusion matrices of both models on the test set, in raw image counts. ConvNeXt-Tiny missed only 1 of the 165 diabetic retinopathy images, compared with 7 for DeiT-Small. For glaucoma, the errors of both models follow the same pattern: the misclassified images are assigned mostly to normal (18 for ConvNeXt-Tiny, 19 for DeiT-Small) and to diabetic retinopathy (14 for both), and only rarely to cataract (7 and 5). In the opposite direction, normal images are most often mistaken for glaucoma (15 and 12), indicating that the glaucoma–normal boundary is the main source of error for both architectures.

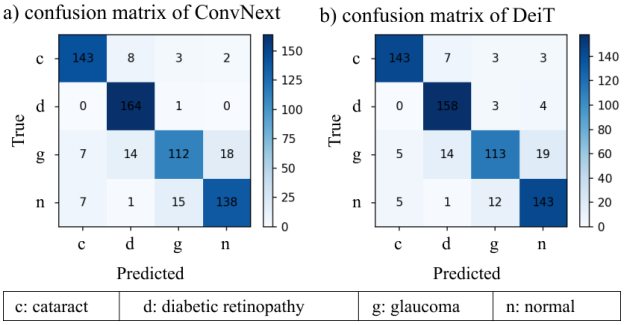


Fig. 4. Confusion matrices of (a) ConvNeXt-Tiny and (b) DeiT-Small on the test set (image counts).

C. Qualitative Explainability Results

Fig. 5 presents qualitative examples of Grad-CAM heatmaps generated from ConvNeXt-Tiny and Attention Rollout heatmaps generated from DeiT-Small on representative test images from each class. For glaucoma and diabetic retinopathy, both methods concentrate their highest activations tightly on the optic disc region – the clinically relevant structure for assessing cupping in glaucoma and a common site of hemorrhages and exudates in diabetic retinopathy – indicating that both models learned to attend to anatomically meaningful evidence rather than spurious background cues for these two classes. For the normal example, both methods likewise center on the optic disc and surrounding vasculature, consistent with this being the reference structure clinicians inspect first. Cataract is the weakest case for both methods: Grad-CAM activates a narrow vertical strip near the image border rather than a diffuse haze pattern (the actual visual signature of a cataract in a fundus photo), while Attention Rollout spreads its attention toward the image corners and edges outside the fundus circle. This is consistent with cataract's presentation as a global loss of image sharpness/contrast rather than a spatially localized lesion, which is inherently harder for patch-level saliency methods – both CNN- and attention-based – to localize. Attention Rollout's tendency to assign non-trivial weight to border/corner patches across multiple classes also matches its characterization in Section III-B as producing more diffuse heatmaps than Grad-CAM.

D. Faithfulness Evaluation

To quantitatively assess the faithfulness of the generated explanations, deletion and insertion AUC were computed for Grad-CAM on ConvNeXt-Tiny and Attention Rollout on DeiT-Small using patch-based perturbation, as described in Section III-C. The scores were computed on all 633 test images for each architecture, each explained with respect to the class predicted by the model.

Both metrics are read from a curve of the model's confidence in its predicted class as the image is perturbed patch by patch in the order given by the heatmap. For deletion, the curve starts at the confidence on the original image and the most important patches are removed first, so a faithful heatmap makes the confidence fall quickly and yields a small area under the curve. For insertion, the curve starts at the confidence on the blurred baseline and the most important patches are restored first, so a faithful heatmap makes the confidence rise quickly and yields a large area. Because both the confidence and the fraction of perturbed patches lie in $[0, 1]$, each AUC also lies in $[0, 1]$ and can be read as the average confidence retained over the whole perturbation sequence. Table IX summarizes the resulting scores, averaged over the test set.

TABLE IX
DELETION AND INSERTION AUC OF GRAD-CAM AND ATTENTION ROLLOUT ($n = 633$ TEST IMAGES)

XAI Method	Deletion AUC ↓	Insertion AUC ↑
Grad-CAM (ConvNeXt-Tiny)	**0.3906**	0.6982
Attention Rollout (DeiT-Small)	0.4626	**0.7263**

A deletion AUC of 0.3906 therefore means that ConvNeXt-Tiny retains on average about 39% confidence in its prediction while the patches ranked highest by Grad-CAM are progressively removed, against about 46% for DeiT-Small under Attention Rollout. Conversely, an insertion AUC of 0.7263 means that DeiT-Small holds on average about 73% confidence while the patches ranked highest by Attention Rollout are progressively restored, against about 70% for ConvNeXt-Tiny under Grad-CAM. The two methods thus split the faithfulness comparison: Grad-CAM is more faithful under deletion (a margin of 0.072), while Attention Rollout is more faithful under insertion (a smaller margin of 0.028). This is consistent with the qualitative pattern in Fig. 5: Grad-CAM's sharper, more concentrated activations identify a small set of patches whose removal quickly lowers the confidence, whereas Attention Rollout's broader coverage is less precise but recovers the confidence slightly faster as patches are restored. Since each method is paired with a different architecture, these scores compare the two model–explanation pairs rather than the two explanation methods in isolation.

TABLE X
PER-CLASS DELETION (DEL.) AND INSERTION (INS.) AUC, GROUPED BY PREDICTED CLASS

	Grad-CAM			Attention Rollout		
Class	n	**Del. ↓**	**Ins. ↑**	n	**Del. ↓**	**Ins. ↑**
Cataract	157	0.687	0.656	153	0.845	0.860
DR	187	0.412	0.892	180	0.550	0.840
Glaucoma	131	0.136	0.692	131	0.089	0.748
Normal	158	0.237	0.527	169	0.291	0.445

Table X breaks the scores down by predicted class. Both methods obtain their lowest deletion AUC on glaucoma (0.136 for Grad-CAM and 0.089 for Attention Rollout), where the pathology is concentrated at a single, well-localized structure (the optic disc), so removing the few highlighted patches is enough to collapse the prediction. Cataract shows the opposite behavior, with the highest deletion AUC for both methods

(0.687 and 0.845): the confidence stays high even after the top-ranked patches are removed, matching the qualitative finding that neither method localizes cataract's diffuse presentation well. A likely contributing factor is the perturbation baseline itself: a heavily blurred fundus image resembles the hazy appearance of a cataract, so replacing patches with blur removes little cataract evidence. The same effect would explain why Attention Rollout reaches a high insertion AUC on cataract (0.860) while both methods obtain their lowest insertion AUC on the normal class (0.527 and 0.445), since a blurred starting image already favors the cataract prediction and many sharp patches must be restored before the model becomes confident that an image is normal.

E. Discussion

Taken together, the classification and explainability results indicate that overall predictive accuracy and explanation faithfulness are not the same axis of comparison, and neither architecture dominates on both. DeiT-Small edges out ConvNeXt-Tiny on test macro-F1 by a negligible margin (87.74% vs. 87.63%), while ConvNeXt-Tiny is clearly preferable on diabetic retinopathy recall (99.39% vs. 95.76%) and macro ROC-AUC (98.15% vs. 97.06%) – and its paired explanation method, Grad-CAM, is also the more faithful of the two under deletion. For a clinical screening system where missing a diabetic retinopathy case is the costlier error and clinicians need tightly localized visual evidence to trust a prediction, ConvNeXt-Tiny with Grad-CAM is the more defensible choice despite its very slightly lower macro-F1. Conversely, DeiT-Small with Attention Rollout attains marginally higher overall macro-F1 and a higher insertion AUC, which may suit contexts that value broader contextual corroboration over pinpoint localization. This reinforces the motivation stated in Section I: accuracy comparison alone is insufficient for selecting a deployable architecture, and faithfulness metrics surface trade-offs that accuracy hides.

This study has several limitations. The hyperparameter search covered only 3 learning rates $\times$ 5 epoch budgets (15 configurations) per architecture, so the reported results reflect the best configuration found within this grid rather than a globally optimal one. The qualitative comparison in Fig. 5 is illustrative (one image per class) rather than an exhaustive audit, and a larger-scale clinical validation with ophthalmologist-annotated lesion masks would be needed to more rigorously confirm the anatomical relevance of the highlighted regions. Finally, both models were fine-tuned and evaluated on a single dataset of $4{,}217$ images from one source; generalization to other fundus imaging devices, populations, or class distributions remains untested.

V. Conclusion

Placeholder conclusion text.

References

[1] World Health Organization, "Blindness and vision impairment," 2023, fact Sheet, Accessed: 2026-08-04. [Online]. Available: https://www.who.int/news-room/fact-sheets/detail/blindness-and-visual-impairment

[2] Vision Loss Expert Group of the Global Burden of Disease Study and GBD 2019 Blindness and Vision Impairment Collaborators, "Global estimates on the number of people blind or visually impaired by cataract: A meta-analysis from 2000 to 2020," *Eye*, vol. 38, no. 11, pp. 2156–2172, 2024.

[3] e. a. Pan, "Global, regional and national burden of blindness and vision loss attributable to diabetic retinopathy, 1990–2021: A systematic analysis for the global burden of disease study 2021," *Diabetes, Obesity and Metabolism*, 2025.

[4] e. a. Vania, "Global incidence and risk factors for glaucoma," *Journal of Global Health*, vol. 14, p. 04252, 2024. [Online]. Available: https://jogh.org/2024/jogh-14-04252

[5] Elkholy and Marzouk, "Deep learning-based classification of eye diseases using convolutional neural network for oct images," *Frontiers in Computer Science*, 2024.

[6] Z. Liu, H. Mao, C.-Y. Wu, C. Feichtenhofer, T. Darrell, and S. Xie, "A convnet for the 2020s," in *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 6 2022, pp. 11 966–11 976.

[7] H. Touvron, M. Cord, M. Douze, F. Massa, A. Sablayrolles, and H. Jégou, "Training data-efficient image transformers & distillation through attention," 2021. [Online]. Available: https://arxiv.org/abs/2012.12877

[8] C. Gomez, T. Fréour, and H. Mouchère, "Metrics for saliency map evaluation of deep learning explanation methods," in *International Conference on Pattern Recognition and Artificial Intelligence (ICPRAI)*, ser. Lecture Notes in Computer Science, 2022. [Online]. Available: https://arxiv.org/abs/2201.13291

[9] e. a. Cho, "Gmar: Gradient-driven multi-head attention rollout for vision transformer interpretability," 2025. [Online]. Available: https://arxiv.org/abs/2504.19414

[10] M. M. Islam, C. K. Shahnaz, and M. A. Hossain, "Eye disease classification dataset," Kaggle Dataset, 2022.

[11] C. Erdas and G. Arslan, "Efficient detection of multiclass eye diseases using deep learning models: A comparative study," *MATTER: International Journal of Science and Technology*, vol. 10, pp. 55–66, 01 2024.

[12] F. Jibon, F. Rabby, N. Rahman, M. Uddin, F. Rushu, M. Islam, M. A. Uddin, A. Khraisat, M. Kazi, and M. A. Talukder, "Tl-med: Multiclass eye disease classification based on ensemble transfer learning and crvo–brvo detection via a single shot multibox detector," *DIGITAL HEALTH*, vol. 11, 09 2025.

[13] M. Colot, C. Simar, M. Petieau, A. M. C. Alvarez, G. Cheron, and G. Bontempi, "Emg subspace alignment and visualization for cross-subject hand gesture classification," *arXiv preprint arXiv:2401.05386*, 2023.

[14] R. Alsohemi and S. Dardouri, "Fundus image-based eye disease detection using efficientnetb3 architecture," *Journal of Imaging*, vol. 11, p. 279, 08 2025.

[15] E.-S. El-kenawy, N. Khodadadi, K. Sh.Gaber, E. Khodadadi, A. Alhussan, D. Khafaga, and M. Eid, "Automated retinal disease classification using deep learning and alexnet with statistical models analysis," *PLOS One*, vol. 21, 01 2026.

[16] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-cam: Visual explanations from deep networks via gradient-based localization," in *2017 IEEE International Conference on Computer Vision (ICCV)*, 2017, pp. 618–626.

[17] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby, "An image is worth 16x16 words: Transformers for image recognition at scale," 2021. [Online]. Available: https://arxiv.org/abs/2010.11929

[18] by Zoltán Varga and U. D. J. Sliwowski, "Multimodal transformer modelsfor human action classification diplom-ingenieur automation and robotic systems (ue066 515)," 2024.

[19] O. Pina, E. Dorca, and V. Vilaplana, "Cell nuclei detection and classification in whole slide images with transformers," *arXiv preprint arXiv:2502.06307v1*, 2 2025.

[20] J. Maurício, I. Domingues, and J. Bernardino, "Comparing vision transformers and convolutional neural networks for image classification: A literature review," *Applied Sciences*, vol. 13, no. 9, p. 5521, 2023.

[21] S. Alayón, J. Hernández, F. J. Fumero, J. F. Sigut, and T. Díaz-Alemán, "Comparison of the performance of convolutional neural networks and vision transformer-based systems for automated glaucoma detection with eye fundus images," *Applied Sciences*, vol. 13, no. 23, p. 12722, 2023.

[22] E. E. Hwang, C. Dake, Y. Han, J. Lin, and J. Shan, "Multi-dataset comparison of vision transformers and convolutional neural networks for detecting glaucomatous optic neuropathy from fundus photographs," *Bioengineering*, vol. 10, no. 11, p. 1266, 2023.

[23] A. R. Chłopowiec, K. Karanowski, T. Skrzypczak, M. Grzesiuk, A. B. Chłopowiec, and M. Tabakov, "Counteracting data bias and class imbalance—towards a useful and reliable retinal disease recognition system," *Diagnostics*, vol. 13, p. 1904, 5 2023.

[24] A. AlMohimeed, "Multi-stage framework using transformer models, feature fusion and ensemble learning for enhancing eye disease classification," *Scientific Reports*, vol. 15, p. 30469, 8 2025.

[25] Y. Bengio, "Continuous optimization of hyper-parameters," in *Proceedings of the IEEE-INNS-ENNS International Joint Conference on Neural Networks. IJCNN 2000. Neural Computing: New Challenges and Perspectives for the New Millennium*. IEEE, 2000, pp. 305–310 vol.1.

[26] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," 2019. [Online]. Available: https://arxiv.org/abs/1711.05101

[27] J. Bergstra and Y. Bengio, "Random search for hyper-parameter optimization," *Journal of Machine Learning Research*, vol. 13, no. 10, pp. 281–305, 2012. [Online]. Available: http://jmlr.org/papers/v13/bergstra12a.html

[28] D. Reyes and J. Sánchez, "Performance of convolutional neural networks for the classification of brain tumors using magnetic resonance imaging," *Heliyon*, vol. 10, p. e25468, 2 2024.

[29] S. Jiang, G. J. Zanazzi, and S. Hassanpour, "Predicting prognosis and idh mutation status for patients with lower-grade gliomas using whole slide images," *Scientific Reports*, vol. 11, p. 16849, 8 2021.

[30] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-cam: Visual explanations from deep networks via gradient-based localization," *International Journal of Computer Vision*, vol. 128, pp. 336–359, 2 2020.

[31] A. Y. Alhafdhi, G. Abosamra, and A. M. Alshareef, "Syncretic grad-cam integrated vit-cnn hybrids with inherent explainability for early thyroid cancer diagnosis from ultrasound," *Diagnostics*, vol. 16, p. 999, 3 2026.

[32] K. Vanitha, T. R. Mahesh, V. V. Kumar, and S. Guluwadi, "Enhanced tuberculosis detection using vision transformers and explainable ai with a grad-cam approach on chest x-rays," *BMC Medical Imaging*, vol. 25, p. 96, 3 2025.

[33] S. A. El-Ghany, M. A. Mahmood, and A. A. A. El-Aziz, "Automated eye disease diagnosis using a 2d cnn with grad-cam: High-accuracy detection of retinal asymmetries for multiclass classification," *Symmetry*, vol. 17, p. 768, 5 2025.

[34] S. Abnar and W. Zuidema, "Quantifying attention flow in transformers," in *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*. Association for Computational Linguistics, 2020, pp. 4190–4197.

[35] Z. Cheng, Y. Wu, Y. Li, L. Cai, and B. Ihnaini, "A comprehensive review of explainable artificial intelligence (xai) in computer vision," *Sensors*, vol. 25, p. 4166, 7 2025.

[36] Y. Li, J. Wang, X. Dai, L. Wang, C.-C. M. Yeh, Y. Zheng, W. Zhang, and K.-L. Ma, "How does attention work in vision transformers? a visual analytics attempt," *IEEE Transactions on Visualization and Computer Graphics*, vol. 29, pp. 2888–2900, 6 2023.

[37] V. Petsiuk, A. Das, and K. Saenko, "Rise: Randomized input sampling for explanation of black-box models," 2018. [Online]. Available: https://arxiv.org/abs/1806.07421

[38] M. Skliarov, R. E. Shawi, C. Dhaoui, and N. Ahmed, "A comparative evaluation of explainability techniques for image data," *Scientific Reports*, vol. 15, p. 41898, 11 2025.

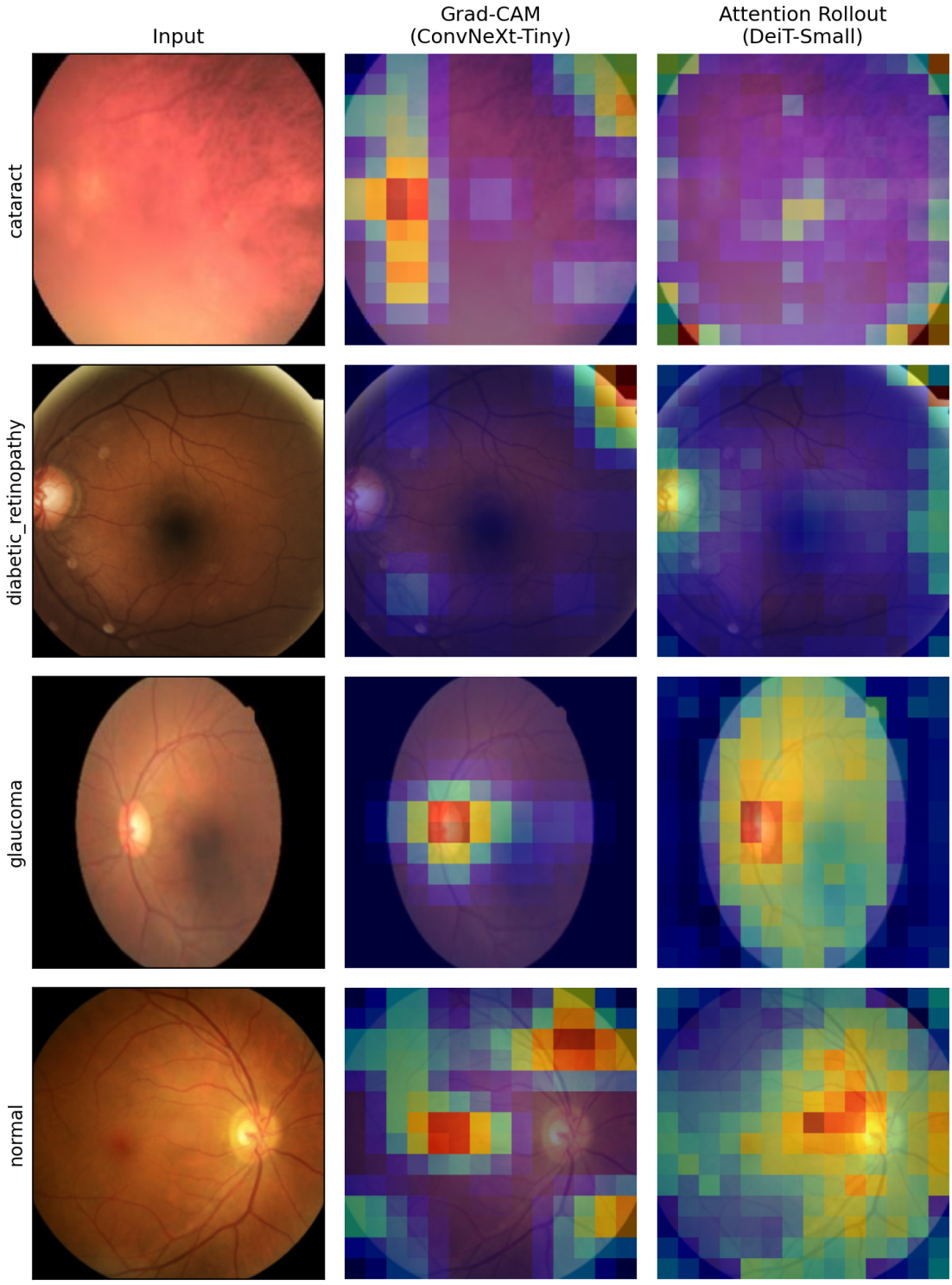


Fig. 5. Qualitative comparison of Grad-CAM (ConvNeXt-Tiny) and Attention Rollout (DeiT-Small) heatmaps across the four eye disease classes.